

Wynn’s World: The Cognitive Architecture at Work in a Minimal Counting Scenario

Abstract

The architecture proposed in [2] (hereafter ‘the architecture paper’) is implemented and tested in a minimal scenario inspired by Wynn’s (1992) infant counting experiments. While both the mechanism and the interpretations of Wynn’s findings remain debated, the experimental setup — controlled perception, known arithmetic, measurable violation — is a tight fit for a focused implementation test of the proposed architecture.

A brute-force matcher searches for productive wirings between an unsupervised perception stream and pre-trained GRU-based predictive models, using a deficit scoring function as the sole evaluation criterion. The implementation produces results that were not designed in: underdetermination of wirings is encountered, not stipulated; the proto-ontology is required, not implemented as a feature; model succession emerges as a rank crossing in deficit tables, not as a scripted transition. We report multiple experimentally confirmed findings, including the distinction between trained and untrained spare capacity, two qualitatively different environmental signals, lateral detection of model inadequacy before new tokens arrive, succession as an inspectable architectural event, and robustness of all findings across scoring functions. All experiments are reproducible from the accompanying repository [3].

1 Introduction

The architecture paper proposes a cognitive architecture built on the independence of perception and prediction, with a matching process as the sole bridge between them. This paper puts that architecture to work.

The immediate question is: is this architecture demonstrable at all? Can one define an experimental setup basic enough to be tractable, but not so basic that the results are trivial?

This is a difficult call. Luckily, we can follow the footsteps of some brilliant behavioral researchers.

In 1992 Karen Wynn was working as a post-doc at the University of Arizona. She designed an experiment methodologically ingenious both in the arrangement and the measurement.

In a dark room a 5-month old infant would be sitting, facing a single bright object: the stage. Behind them, out of sight, their parents would be present, providing the feeling of security, while not contaminating the infant’s perception

with social interactions. On stage, equipped with a rotating screen, Mickey Mouse dolls would appear and disappear, going behind and coming out from behind the screen. Sometimes the screen is moved away, revealing how many dolls are at this moment behind it.

First the experiment goes through a familiarization phase. The world does not cheat: if two dolls have gone behind the screen and the screen goes up, two dolls are revealed. If two dolls were behind the screen, and one of them has been seen going away, removing the screen reveals one doll, as expected.

Then the world begins to — occasionally — cheat. The count of the dolls behind the screen defies the rules of arithmetic.

The measurement is amazingly simple: it is the temporal trace of infant’s attention, or, in plain words, how long it looks at the revealed dolls. The correlation between the attention given and the arithmetical mismatch is the experiment result: five-month-olds, who can’t speak, can’t point, can barely move, demonstrate through looking time alone that something in their head expected a different count.

We are going to adopt this setup almost unchanged: the unmodulated perception pipeline, the perception signals relating only to dolls arrivals, departures, and their observed count. The only adaptations we are making are increasing the number of dolls to better register the interesting phenomena within the architecture.

This is a companion paper. Its goal is not to demonstrate cognitive realism. The goal is to show that the architecture presented in [2], taken seriously as an implementation constraint, forces outcomes that the architecture paper claims. Where the architecture paper argues, this paper exhibits.

2 The Minimal World

World dynamics. Perception needs some external observables to work on. Following Wynn’s elimination and focus, the “world” in our experiment has two events and one state, corresponding to doll arrival, doll departure, and dolls count, respectively.

More precisely: the world maintains an integer count state. At each step, one of three things happens: an increment (INC), a decrement (DEC), or a read of the current count (C0, C1, C2, ...). The world deals with a limited number of dolls available. If there are three dolls available, the possible count readouts are (C0, C1, C2, C3). We call such a world a “capacity 4 world” or, in short, cap4 world.

The world is realistic: the observed number of dolls never turns negative, nor does it surpass the total number of dolls available. DEC cannot happen if the current count is C0; INC is not possible if the count is at the maximum.

Otherwise the world works as expected, emitting an INC event when increasing the state, and a DEC upon decreasing it.

Following the correspondence to the real world setup, the “capacity” of the world is a parameter that can be changed: another doll can be introduced, moving this limit of the world and making another readout available.

A cap3 world has count states $\{0, 1, 2\}$ and token vocabulary $\{\text{INC}, \text{DEC}, \text{C0}, \text{C1}, \text{C2}\}$. A cap4 world has count states $\{0, 1, 2, 3\}$ and token vocabulary $\{\text{INC}, \text{DEC}, \text{C0}, \text{C1}, \text{C2}, \text{C3}\}$. A cap5 world extends further.

The infant in the dark room observes the scene. If it observes a doll going in — we model this as an INC perception token. A doll goes out — a DEC. Observing dolls on the scene results in different tokens C0, C1, ..., depending on the dolls count. These are not “counts” — they are just distinct tokens. We label them the way we do because *we* know. The infant does not. At least not initially.

There is nothing more in this world. The infant is concerned only with tracking numerosity of a single category.

3 Implementation

Boxed predictive models. The architecture calls for perception to be contrasted with and played against a portfolio of predictive automata. In our implementation each automaton is a GRU-based recurrent network, pre-trained in isolation on sequences from its training world. The model exposes a fixed-size token dictionary and returns probability distributions over next tokens. It is important that the dictionary and the probability distribution are exposed uniformly: even if internally some tokens correspond to state transitions, and others to state readouts, this distinction is not published in any way outside the model. The matcher sees only the dictionary size and the output distributions.

The full encapsulation of the internal predictive logic of a model allows for inclusion of a fully deterministic symbolic counter as an interesting benchmark and a useful sanity check.

Curriculum training. Models are trained in stages: cap1→cap2→cap3→cap4, each stage consolidating before the next. Without curriculum, the GRU achieves similar accuracy but with internally scrambled count state representations.

Three model variants are used: “native,” with full curriculum pre-training; “ghost,” with pre-training stopped early; and “symbolic” — a deterministic benchmark.

Most of the actual experiments use three models:

Model	Architecture	Training	Purpose
acc70	cap4 GRU	Full curriculum, ~70% accuracy	Lower-quality baseline
acc80	cap4 GRU	Full curriculum, ~80% accuracy	Primary experimental model
ghost	cap4 GRU (or cap5)	Curriculum stopped early	Tests adaptive potential

The matcher. Following the architecture paper, the system must find the wirings between the perception stream and the predictive models that result in low predictive error. In this experiment this is done brute-force / incremental: initially the matcher evaluates every possible wiring (permutation of perception tokens to model tokens) against the perception stream. Each wiring is scored by deficit: the cumulative shortfall between the model’s predicted probability for the actual next token and the uniform baseline. Upon expansions of the working dictionaries, the matching process evaluates possible extensions of its current portfolio of wirings.

The matcher’s internal state — the structure that makes it possible to ask “what if perception token X maps to model token Y?” before any mapping exists

— is what the architecture paper calls the proto-ontology. In this implementation, it is the mapping vector between perception tokens and model tokens: initially unassigned, populated by the matching process, and retained as the fossilized wiring. The matcher is stateful not only for the sake of the matching process: retaining the predictively performant wirings is necessary both for exploitation and for the incremental adaptation.

The brute-force search is feasible here because the world is small. It is not a requirement of the architecture — it is a choice that makes the matcher’s behavior fully inspectable.

Deficit and the renormalized distribution. The predictive deficits of the available wirings must be calculated in order to identify the best ones. The architecture paper calls for a strict isolation between the internal machineries of the perception, the prediction, and the matcher. This makes the design of the deficit function an interesting exercise. The function must be oblivious to the internal ontologies of the models, should be independent of history, and should be simple.

Using the uniform distribution over the model’s token dictionary as a baseline, we define the predictive error at any point as $\max(1/n_{\text{tokens}} - p_{\text{pred}}(\text{actual_next}), 0)$, and the predictive deficit as the error summed over steps, averaged over sequences. A wiring that predicts no better than chance accumulates maximum deficit; a wiring whose model confidently predicts the correct next token accumulates zero.

The choice of this specific deficit formula affects what is visible but not what is true. As shown in Appendix Appendix A, the architectural findings hold across scorer families, though different scorers expose different aspects of the deficit landscape.

Cliff pruning. After scoring all wirings, contenders are selected by finding the largest gap in the sorted deficit list. Everything below the gap is retained. This is adaptive — it reads the structure of the data rather than imposing a fixed threshold. The contender count is the matcher’s epistemic humility about which wiring will generalize.

Incremental wiring. When the world expands (e.g., $\text{cap3} \rightarrow \text{cap4}$), the matcher does not restart from scratch. It extends each surviving contender by searching only over the unassigned model tokens for the new perception token. Fossilized wirings constrain the search space.

Experimental conditions. Unless otherwise noted: $n_{\text{seeds}}=5$, $\text{seq_len}=300$, $\text{seed}=42$, $p_{\text{read}}=0.6$.

4 The Matcher Finds Correct Wirings

We begin the experiment with the world that contains two dolls. At most two dolls are visible at any moment. Three dolls visible are not possible — there are not that many dolls in this world. The world does not cheat: the arithmetic is truthful, the number of dolls does not decrease below zero nor does it increase over two.

The infant tries to align its predictive potential with perception. In Wynn’s terms this corresponds to *familiarization*.

The matching process arrives at the following ranking of the possible wirings:

Place	Deficit	Model	INC	DEC	C0	C1	C2
1	0.0000	cap4_ghost	INC	DEC	C0	C1	C2
2	0.0102	cap3_acc70	INC	DEC	C0	C1	C2
3	0.0136	cap4_ghost	DEC	INC	C2	C1	C0
4	0.0287	cap3_acc80	DEC	INC	C2	C1	C0
5	0.0573	cap4_acc80	DEC	INC	C2	C1	C0
6	0.0712	cap3_acc70	DEC	INC	C2	C1	C0
7	0.0789	cap3_acc80	INC	DEC	C0	C1	C2
8	0.2405	cap4_acc80	INC	DEC	C0	C1	C2
9	0.3025	cap4_acc80	INC	DEC	C0	C1	C3
10	0.3857	cap4_acc80	DEC	INC	C3	C1	C0
11	1.5831	cap3_acc70	C1	DEC	C0	INC	C2
12	1.6048	cap4_acc80	INC	DEC	C1	C2	C3

There are three things to observe about this table.

The Cliffs. We have truncated it at the twelfth position to make the cliffs visible. The first cliff is between position 7 and position 8. The second cliff, between positions 10 and 11, is an order of magnitude steeper. Any wiring up to the first cliff is a reasonable contender. Any wiring beyond the second cliff can be discarded. The fate of the wirings in-between depends on the system’s appetite for adaptability.

The Symmetry. There are two families of wirings which perform well: the “direct” wirings, and “mirror-image” wirings. The direct wirings are what we expected: the world’s INC event is mapped to the model’s INC, DEC to DEC, and the numerosities are mapped directly. The mirror-image wirings are only obvious in hindsight: the world has a reflection symmetry which the mapper faithfully reports. After truncating the contenders list at the cliff, the system is holding two contradictory-but-equal bets about where the world’s count axis sits in the model’s vocabulary.

Wiring family	INC→	DEC→	C0→	C1→	C2→	Best deficit
Canonical	INC	DEC	C0	C1	C2	0.0000
Reverse-canonical	DEC	INC	C2	C1	C0	0.0136

In the context of architecture paper claims: the matcher does not care for nor is aware of the internal logic of a model. Operational performance is all the signal. The architectural principle has determined what has unfolded.

The Ghost. The best performing wiring connects to the model with an *untrained* C3 pad. The perfect performance of this model in the cap2 world is a surprise, and can probably be attributed to its internal spare capacity. As demonstrated in the next section, however, betting solely on this model in a world that may evolve would be perilous.

Note on the deficit function. The specific formula we have chosen involves a very simple renormalization of the returned probability distribution, which depends on the wired dictionary size. Without renormalization, a model that spreads mass across unwired pads would appear worse than it is. This is the only place in the architecture where any calibration of a “capN model” to a “capM world” happens.

5 The Arrival of the Third Doll

Incremental familiarization. It had taken approximately five months for the infants from Wynn’s experiment to arrive at a developmental stage where the mismatch on counts up to 2 would trigger the observed behavior. We have no idea how much more time it would take them to display similar behavior with up to 3 dolls. For us, however, this is a very natural next step in the experiment.

When the world expands incrementally (cap3→cap4→cap5), fossilized wirings from the previous stage constrain the search at the next stage. The effect is dramatic.

Stage	World	Evaluations per model	Brute-force equivalent
Stage 1	cap3 (5 tokens)	$P(7, 5) = 2,520$	—
Stage 2	cap4 (6 tokens)	$9 \times 2 = 18$ contenders	$P(7, 6) = 5,040$
Stage 3	cap5 (7 tokens)	$14 \times 1 = 14$ contenders	$P(7, 7) = 5,040$

At stage 2, instead of evaluating 5,040 wirings from scratch, the matcher extends 9 surviving contenders over 2 remaining model tokens: 18 evaluations. At stage 3, 14 extensions with 1 remaining token each. The fossilized wirings are not just records — they are search constraints. This is the computational consequence of fossilization that the architecture paper describes as “the system’s ontological commitments.”

The incremental strategy works only as long as the final contenders survive the pruning at the earlier stages. We found that our simple “cliff” heuristic performs well. We do not go into detailed study of the cutoff regimes to keep this paper focused.

Adaptive potential. We model the arrival of the third doll — the evolution of the world — as a simple change to the perception token sequences: we include C3 as a new token in the perception vocabulary, and we move the constraint on the appearance of INC from C2 to C3 (allowing for C2→C3 transitions and forbidding increases beyond C3). We run the matching competition on those contenders from the cap3 pool which have what the architecture paper calls adaptive potential (i.e., at least one unwired pad).

In this pool we have a “native” model, pre-trained to count up to three, and a “ghost” model. The ghost model has the same architecture as a native model (same number of tokens, same GRU) but its curriculum was stopped early. A cap4 ghost trained only to cap3 has the C3 slot in its vocabulary, but C3 received zero gradient during training. The slot exists architecturally but carries no representation. Formally, until tested against the cap4 world, both models promise the same adaptive potential.

On the home world (cap3), the ghost wins:

Place	Deficit	Model	Wiring
1	0.0000	cap4_ghost	canonical
2	0.0136	cap4_ghost	reverse-canonical
3	0.0577	cap4_acc80	reverse-canonical
4	0.3795	cap4_acc80	canonical

The ghost’s relaxed interior representation (cap4 architecture trained to cap3 boundary) outperforms native cap3 models on their own world.

On the extended world (cap4), the ghost collapses:

Place	Deficit	S1 Base	Model	Wiring
1	0.0400	0.3795	cap4_acc80	canonical
2	3.1905	0.5072	cap4_acc80	shifted
3	7.0694	0.0000	cap4_ghost	canonical
4	8.6570	0.0136	cap4_ghost	reverse-canonical

The ghost’s leading wiring deficit goes from 0 to over 7. The ghost’s C3 slot is inert. No valid wiring exists — all ghost wiring attempts on cap4 score ≥ 6.36 .

We see a dramatic example of what the architecture paper calls wiring and model succession. The value of retaining contenders is demonstrated: the correct path forward was not the best current wiring but a weaker contender with the right base for extension.

Wiring succession (to the same model). The architecture paper demands that we demonstrate succession between competing wirings to the same model. This is clearly visible in the scores of the cap4_acc80 model against cap3 and cap4 worlds:

2-Dolls World

Deficit	Model	INC	DEC	C0	C1	C2
0.0573	cap4_acc80	DEC	INC	C2	C1	C0
0.2405	cap4_acc80	INC	DEC	C0	C1	C2

3-Dolls World

Deficit	Model	INC	DEC	C0	C1	C2	C3
0.0400	cap4_acc80	INC	DEC	C0	C1	C2	C3
0.9823	cap4_acc80	DEC	INC	C3	C2	C1	C0

The mirror-image wiring performs better in the smaller world, but collapses when the world extends. The model remains, but a different wiring takes the lead.

6 Two Kinds of Environmental Change

Intuitively, there is a difference for an infant between observing something *new* and observing something *wrong*. The very reason for the familiarization phase in Wynn’s experiment is to remove the *new* and isolate the *wrong*. In our simple setup, this intuition becomes tractable. The architecture paper describes three signals ruling the system’s mechanics: the predictive error, the adaptive potential, and the presence of unwired perception tokens (which the paper calls the “cognitive itch”).

Let us narrow our focus to the two models from the previous sections: the “native” cap4_acc80, and the cap4_ghost. The cap3 world has been “familiarized”: optimal

wirings to both models have been identified and fossilized (see the score table in Section 4).

In the previous section we have seen the outcome of the re-wiring effort. Now let us inspect in some detail the ontological crisis as it unfolds, but before it is overcome. What are the signals available when the world changes, which could guide the adaptation?

The New. The infant has familiarized the world of two dolls. The parent, or the nice researcher, has brought another doll. The world, as reported by the perception stream, offers new phenomena: there is now a C3 perception token available, and INC can happen when the count state is 2. How do the wirings fossilized at cap3 perform?

It turns out that even if the new token is filtered from the evaluation stream, the ghost still fails. We verified this directly: filtering all C3 tokens from the evaluation stream, the ghost still fails with a deficit of 3.3845 — nearly half of its unfiltered deficit of 7.0694. The boundary dynamics alone, without any new token present, are sufficient to expose model inadequacy. Where the architecture paper talks about “blindness” to the arrival of new tokens, we identify a “lateral detection” of the world change. Its specific form is determined by the nature of our experimental setup, but in general we expect this phenomenon to stem from the predictive nature of the models and to occur often.

A simple simulation shows how the prediction deficits change when the world extends, but the wirings remain:

Place	Old Deficit	New Deficit	Model	INC	DEC	C0	C1
1	0.3795	0.0400	cap4_acc80	INC	DEC	C0	C1
2	0.0577	10.3050	cap4_acc80	DEC	INC	C3	C2
3	0.0000	7.0694	cap4_ghost	INC	DEC	C0	C1

The one wiring of the native counter performs well, even significantly improving its performance — it correctly predicts the INC/DEC dynamics at the new boundary without observing the intermediate count state. The additional world structure disambiguates this wiring. The inverse wiring collapses. The ghost counter collapses too — it has never seen INC events happening when the count is C2.

Decomposing the deficit by predicted token type reveals a directed gradient:

Wiring	INC	DEC	C0	C1	C2	C3	Total
ghost canonical	0.0000	0.0000	0.0000	0.0300	3.3545	3.6849	7.0694
/step	0.0000	0.0000	0.0000	0.0001	0.0120	0.0297	
ghost reverse	0.0000	0.0000	0.0136	0.2802	4.3211	4.0422	8.6570
/step	0.0000	0.0000	0.0001	0.0008	0.0154	0.0326	
acc80 canonical	0.0000	0.0000	0.0000	0.0400	0.0000	0.0000	0.0400
/step	0.0000	0.0000	0.0000	0.0001	0.0000	0.0000	
acc80 reverse	0.0000	0.0000	0.0549	1.1345	5.0430	4.0725	10.3050
/step	0.0000	0.0000	0.0004	0.0031	0.0180	0.0328	

The results are telling: the error accumulates at the C2/C3 boundary, except for the one wiring of the native model that happens to be aligned with the extension.

The other wiring of the native model performs the worst. The ghost model fails in a more symmetric fashion. It tolerates orientation reversal because it has nothing at C3 to lose; the native model’s wirings split because it does.

The lateral detection provides a structured fingerprint of the world change, and is accompanied by the “itch” of the arrival of a new, unwired perception token. It turns out that the infant has surprisingly rich data prior to initiating the search for productive extensions of its fossilized wirings.

The Wrong. Let us now consider the situation where the world starts “cheating.” There is nothing new in the world or in the perception stream, but 10% of the world read events report the wrong count (C_k replaced by a random C_j , $j \neq k$). The ranking of the prediction deficit of the wirings in the portfolio changes:

Place	S1 Def	Viol Def	Model	INC	DEC	C0	C1	C2
1	0.0136	1.1062	cap4_ghost	DEC	INC	C2	C1	C0
2	0.0000	1.1137	cap4_ghost	INC	DEC	C0	C1	C2
3	0.0577	1.3079	cap4_acc80	DEC	INC	C2	C1	C0
4	0.3795	1.6938	cap4_acc80	INC	DEC	C0	C1	C2

In contrast with the world extension, now all wirings’ predictive scores degrade. There are no obvious candidates for a wiring extension. Moreover, there are no new tokens in the perception stream. For our simple system the only way to improve the predictive performance is to reset the predictive portfolio and re-bootstrap. A system with a more elaborate perception pipeline might try to focus on identifying any missing perception details. This is, in our picture, the origin of the “heightened attention” of the infants in Wynn’s experiment — something more basic than “surprise.” Under the experiment’s conditions there is no productive adaptation path available. What remains is re-processing.

The distinction. The world extension and the screen violation are therefore distinguishable not just in magnitude but in architecture: extension produces a directed signal that concentrates at the training boundary and discriminates in favour of the model with trained capacity. Violation produces a uniform, inverted signal that carries no information about which model should succeed. The signatures are very different. The baby can in principle distinguish between seeing something new and being fooled.

Note: these patterns repeat themselves when the parent brings yet another doll — as we have tested for the cap4→cap5 transition.

7 Discussion

The original purpose of the paper was to provide a toy model of the cognitive architecture described in [2]. Modelling the streamlined world of Wynn’s experiment has allowed us to capture the architecture in a very mechanistic form. Experimenting with actual implementations, however simple, brought to light a few new aspects, and refined our views — still staying truthful to the architecture paper. The architecture paper argues that certain outcomes must follow from certain principles. This paper puts those principles to work and reports what happened.

The underdetermination wasn't stipulated — it appeared in the deficit table at stage 1, where the ghost and its mirror image both scored near zero. The succession wasn't programmed — it fell out of comparing two ranked deficit lists across stages. The findings are the architecture paper's claims; the implementation is their witness. Appendix Appendix A contains a full list of observations and findings that had not been pre-programmed by the architecture paper.

What the implementation forced.

The proto-ontology. The matching process requires a working state: a data structure that records, for each perception token, which model token it is currently mapped to. This was not designed in as a named architectural component. It was required because the incremental wiring strategy needed somewhere to record which assignments were fossilized. The architecture paper argues that proto-ontology is a requirement of any perception-prediction coupling architecture. The implementation confirms: there is no way to implement incremental wiring without something that plays this role. Without retaining the top wirings fossilized in the proto-ontology (i.e., the matcher's state), the whole matching process would be pointless, and incremental adaptation impossible.

The interface principle is confirmed as a functional requirement: when violated, the matching problem becomes unsolvable (Appendix Appendix A, Finding 13).

The ghost result. The architecture paper distinguishes architectural capacity from trained capacity. The ghost has a slot for C3 but no trained representation behind it. This distinction was expected. What the implementation reveals is the magnitude: a deficit gap of 7.03 between ghost and native on the extended world, while the gap on the home world runs in the other direction. The numerical gap is not a design choice — it reflects how severely the GRU's hidden state is corrupted when an untrained slot is forced into active use. The architecture paper describes this as a prediction-driven discrimination signal. The 7.03 is that signal's amplitude.

Lateral detection. This was not predicted by the architecture paper and was not a design goal. The implementation produced it as a side effect of routing all signals through a single deficit scorer. The ghost's failure is visible before the new token appears because the INC/DEC dynamics at the training boundary are already wrong. As described in Section 6, this was discovered by running the experiment, not by reasoning about it in advance. The architecture paper describes the cognitive itch (Signal 2) as the primary trigger for wiring search. The implementation adds a prior signal — predictive deficit degradation appears before the cognitive itch. The architecture's self-diagnosis is earlier than the architecture paper describes.

What remains open.

The world is minimal. Three to five count states, GRU-based models, brute-force search. The architecture paper claims at Marr's computational level; this paper exhibits at the algorithmic and implementation levels for one scenario. Whether the findings generalize to richer worlds, larger models, and approximate matching is an empirical question.

Whether the reproducible findings in a minimal world constitute theory confirmation or illustration is the appropriate skeptical question. The architecture paper's own criterion is that the architecture should make observable predictions in implemented systems. These findings are that observation. The reader is in the

better position to judge whether the threshold has been crossed.

The brute-force matcher is the control condition. It establishes what the architecture can find. Approximate matchers are the scaling question, but you cannot evaluate them without first knowing what the correct answer is.

8 Conclusion

The implementation forces things. The proto-ontology is forced by the need for an incremental state. The contender pool is forced by the mathematics of world extension — the correct model cannot always be the current leader. The succession is forced by the same scoring function that tracked performance on the home world. Lateral detection is forced by the GRU’s boundary dynamics. None of these were designed. They were encountered.

The architecture paper argues these outcomes follow from three independence principles: perception doesn’t know the prediction vocabulary, prediction doesn’t know the perception stream, and matching evaluates the coupling without access to either interior. This paper implements those principles and reports what follows. The cliff heuristic embodies the specific epistemic stance the architecture requires: intermediate scores are unreliable discriminators, and preserving minority contenders is the cost of remaining open to a world that has not yet finished changing.

The code and all experiments are available in the accompanying repository [3].

Appendix A Complete Findings Register

The following findings emerged from the implementation. Items marked [body] are presented in the main text; items marked [appendix] are documented here with supporting data. All are reproducible from the repository.

Architectural findings

F1. The matcher recovers correct wirings. Brute-force search over all permutations identifies the two structurally valid wiring families (canonical and reverse-canonical) across all models and seeds. [body, Section 4]

F2. Underdetermination is encountered, not stipulated. The two valid families reflect the counting world’s reflection symmetry. The matcher reports this faithfully — it was not designed to find symmetry. [body, Section 4]

F3. The deficit landscape has cliff structure. Wirings cluster into quality tiers separated by sharp gaps. The cliff between valid and invalid wirings is an order of magnitude steeper than the gap between valid families. [body, Section 4]

F4. Fossilization collapses the search space. Incremental wiring reduces stage-2 search from $P(7, 6) = 5,040$ evaluations to 18, and stage-3 from $P(7, 7) = 5,040$ to 14. Fossilized wirings are search constraints, not just records. [body, Section 5]

F5. Architectural spare capacity without trained knowledge is not adaptive potential. The ghost model (same architecture, curriculum stopped early)

wins on the home world but collapses on the extended world with a deficit gap of 7.03. The slot exists but carries no representation. Replicated at cap5 scale. [body, Section 5]

F6. Model succession is an inspectable rank crossing. The ghost leads at stage 1 (deficit 0.0000); the native model leads at stage 2 (deficit 0.0400). The crossing is visible in the deficit tables without additional diagnostics. [body, Section 5]

F7. Wiring succession occurs within the same model. The reverse-canonical wiring of cap4_acc80 outperforms its canonical wiring on the cap3 world (0.0577 vs 0.2405). On the cap4 world, the canonical wiring wins (0.0400 vs 0.9823). The model remains; the wiring changes. [body, Section 5]

F8. The correct successor was not the stage-1 leader. The acc80 canonical wiring that wins stage 2 (deficit 0.0400) had stage-1 deficit 0.3795 (rank 4). Committing to the stage-1 best would have yielded deficit 10.31 at stage 2. [body, Section 5]

Signal structure findings

F9. World extension produces a directed, boundary-concentrated signal. The ghost’s deficit concentrates at C2 and C3 (98.9% of total). The contamination gradient originates at the untrained slot and decays with distance from the boundary. Replicated at cap5 (C3/C4 boundary). [body, Section 6]

F10. Screen violation produces a uniform, inverted signal. Under 10% count corruption, all wirings degrade. The ghost outperforms the native (1.11 vs 1.69). Selecting on violation signal would promote the wrong model. [body, Section 6]

F11. The inversion holds across all contenders. Scoring all 18 stage-1 wirings under violation, ghost wirings dominate the top ranks. The native acc80 canonical wiring — the one that wins world extension — ranks 7th. The pattern replicates at cap5. [appendix]

F12. Lateral detection: model inadequacy is detectable before new tokens arrive. Filtering C3 from the evaluation stream, the ghost still fails (deficit 3.3845 vs native’s 0.0400). Boundary dynamics leak through INC/DEC before the explicit new token appears. [body, Section 6]

Implementation findings

F13. The interface principle is a functional requirement. Same model weights, two modes: bypass_on scores 0.0518; bypass_off scores 6.0289 with scrambled wiring. Violating the interface makes matching unsolvable. [appendix]

Mode	Best deficit	Best wiring
bypass_on	0.0518	canonical
bypass_off	6.0289	scrambled

F14. Repeg degrades matching monotonically. Re-pegging the GRU hidden state toward trained count attractors during matching hurts performance at every tested alpha value ($0.0 \rightarrow 0.1 \rightarrow 0.2 \rightarrow 0.5$: deficit $0.05 \rightarrow 1.38 \rightarrow 2.27 \rightarrow 3.25$). The model’s internal coordinate system is its own business. [appendix]

F15. Curriculum training is necessary for clean internal geometry. Without staged curriculum, the GRU achieves similar accuracy ($\sim 80\%$) but with internally scrambled count state embeddings. The deficit landscape becomes flatter and less discriminating. [appendix]

F16. All findings are robust across scoring functions. Ghost failure ordering and succession crossing hold under three scorers: deficit (hinge), tiered (stepped penalty), and cross-entropy (NLL). The succession event is in the structure, not in the scorer. [appendix]

Model	deficit	tiered	cross_entropy
symbolic	0.0000	0.0000	283.7
native	0.0400	1.80	382.1
ghost	7.0694	219.2	537.9

Scorer	Stage-1 leader	Stage-2 leader
deficit	ghost (0.0000)	native (0.0400)
tiered	ghost (0.0000)	native (1.80)
cross_entropy	ghost (374.0)	native (382.1)

Cross-entropy reveals what the deficit threshold hides: the symbolic model (283.7) and native model (382.1) are distinguishable, and wrong wirings are punished exponentially (871–998 vs 4.33–5.27 under deficit).

F17. Model quality affects discrimination sharpness. acc70 produces 8 stage-1 contenders vs acc80’s 4. Lower quality yields a flatter deficit landscape with smaller gaps. acc70’s best stage-1 wiring is shifted (non-canonical) — a different training seed broke the symmetry differently. [appendix]

F18. The contender pool is an existential requirement. Four pruning policies tested on cap5 three-stage succession: [appendix]

Policy	Native contenders	Ghost contenders	Final winner	Final deficit
threshold=0.05	0	1	ghost	2.8887
cliff (current)	2	12	native	0.0391
threshold=2.0	4	4	native	0.0391
topk=1	1	0	native†	18.4069

Aggressive pruning (threshold=0.05) eliminates the correct model before the decisive stage. Greedy commitment (topk=1) locks onto a path that yields catastrophic deficit. The cliff policy succeeds because it reads the distribution rather than imposing a fixed threshold. The contender set is the architecture’s hedge against world changes it has not yet observed.

Appendix B Experimental Conditions

All experiments: $n_seeds=5$, $seq_len=300$, $seed=42$, $p_read=0.6$. Deficit scorer: $\max(1/n_{tokens} - p_{pred}(\text{actual_next}), 0)$ summed over steps, averaged over sequences.

Experiment	Arena	Models	World
Underdetermination	cap3_ghost_games	3×cap3, ghost, cap4_acc80	cap3
GRU bypass	cap4_bypass_compare	cap4_acc80 (two modes)	cap4
Repeg sweep	cap4_repeg_sweep	cap4_acc80 (4 alphas), symbolic	cap4
Ghost (cap4)	cap4_ghost_games	ghost, native, symbolic	cap4
Ghost (cap5)	cap5_ghost_games	cap5_ghost, cap5_acc70, cap5_acc80	cap5
Error structure	cap4_error_structure	ghost, native	cap4 (A) / cap3+corrupt (B)
Error structure	cap5_error_structure	ghost, native	cap5 (A) / cap4+corrupt (B)
Lateral detection	cap4_lateral_detection	ghost, native	cap4 (filtered/unfiltered)
Succession (2-stage)	cap4_succession	ghost, cap4_acc80	cap3→cap4
Succession (3-stage)	cap5_incremental	ghost, cap5_acc70, cap5_acc80	cap3→cap4→cap5
Model quality	cap4_incremental_topk	ghost, acc70, acc80	cap3→cap4
Scorer robustness	cap4_ghost_games_scorers	ghost, native, symbolic	cap4
Scorer robustness	cap4_succession_scorers	ghost, cap4_acc80	cap3→cap4

Full deficit tables for all experiments are available in the repository.

Appendix C Model Specifications

All learned models share the same architecture: token embedding dimension 8, GRU hidden dimension 16, one GRU layer. Token vocabulary: INC (0), DEC (1), $C_0 \dots C_{n-1}$ ($2 \dots n+1$), giving $n_{tokens} = n_{counts} + 2$.

Model	n_{tok}	n_{cnt}	Emb	GRU	Training	Accuracy
cap3_acc70	5	3	8	16	Full cap3 curriculum	~70%
cap3_acc80	5	3	8	16	Full cap3 curriculum	~80%
cap4_acc70	6	4	8	16	Full cap4 curriculum	~70%
cap4_acc80	6	4	8	16	Full cap4 curriculum	~80%
cap4_ghost	6	4	8	16	Cap3 curriculum only	~80% on cap3
cap5_acc70	7	5	8	16	Full cap5 curriculum	~70%
cap5_acc80	7	5	8	16	Full cap5 curriculum	~80%
cap5_cap4trained_acc80	7	5	8	16	Cap4 curriculum only	~80% on cap4
symbolic_cap4	6	4	—	—	By construction	100%

References

- [1] Wynn, K. (1992). Addition and subtraction by human infants. *Nature*, 358, 749–750.
- [2] [Architecture paper — arXiv ID to be assigned at simultaneous submission]
- [3] Repository: https://github.com/kubasiemion/cognitive-architecture/tree/main/demos/wynns_world